



Elephant Agent: Agency-First Personal AI

Personal-Model-First Evolution for Judgment, Evidence, Questions, and Growth

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Abstract. Many agent systems now improve along three visible axes: executing tasks, carrying context, and improving procedures. For personal AI, these are not the final product position. The fourth level is agency-first: the system should help the person keep judgment, evidence, questions, and growth while agents do more work. Elephant Agent is a *personal-model-first* architecture for this fourth level of personal AI. It centers a Personal Model: a durable, grounded understanding across four lenses: who the user is (*Identity*), what surrounds them (*World*), what is alive right now (*Pulse*), and what their path has taught them (*Journey*). This system treats memory as five cooperating layers rather than a transcript archive or a vector store: Matriarch Core / Personal Model, Elephant State, Episode / Loop / Step Trail, Contextual Recall, and Background Learning. Raw Steps preserve what happened; contextual recall retrieves support from Steps and Facts; proactive questions expose gaps, conflicts, stale Pulse, and useful adaptations; background learning decides when evidence should change durable claims. The system’s self-evolution loop works as follows: curiosity identifies what would improve future help, and reflect jobs turn lived evidence into governed Personal Model updates. There is no free-form memory-note table or foreground memory write surface. Durable understanding is written through governed Personal Model tools and remains inspectable, correctable, and source-provenanced. In this report, we describe the agency-first product position, Elephant Agent architecture, four-lens Personal Model schema, proactive curiosity loop, local semantic recall path, background reflect pipeline, dashboard inspection surfaces, and evaluation scenarios for wake recovery, correction, no-match recall, and memory quality over time.

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1 Introduction

Elephant Agent starts from the animal before the architecture. Elephant memory is not merely large storage. Elephants recognize herd members, remember danger cues, return to important locations after long gaps, and preserve bonds with other animals and humans over years. Their hippocampus links emotion to long-term memory, while their cerebral cortex supports problem solving, cooperation, tool use, and simple quantitative tracking. In older matriarchs, memory can become practical judgment for the herd: a remembered drought, route, or warning sign changes future behavior.

Personal AI has an analogous design problem. It is not only an assistant that can answer the current prompt. It is an individual system that should become better at working with a particular person across days, surfaces, interruptions, corrections, and repeated work. The product position is agency-first: personal AI should help the user do more without thinking less. The central improvement target is therefore not a larger chat window, a longer log, a vector memory store, or a growing catalog of reusable skills. The central target is the Personal Model: the explicit object that says what the agent currently

Personal-Model-First Memory for Personal AI

Experience becomes governed understanding, not a larger memory pile.

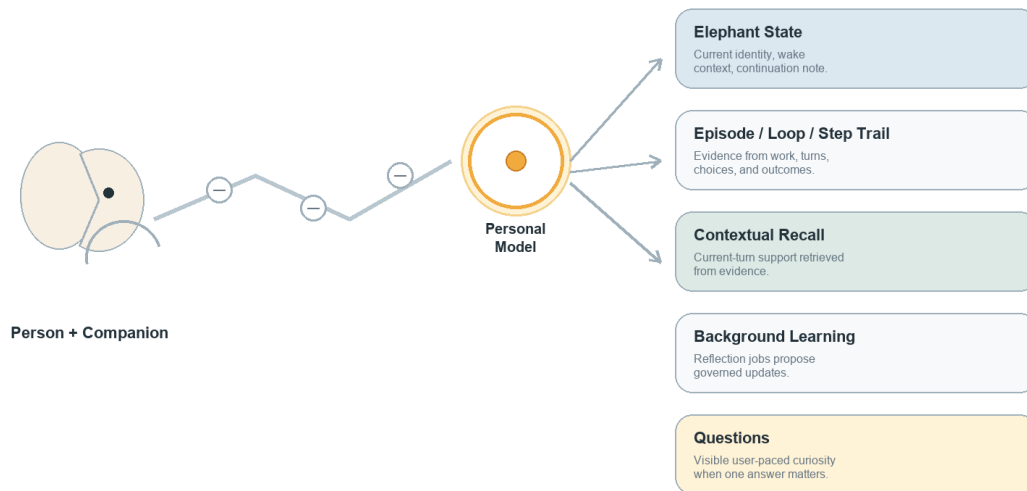


Figure 1 Elephant Agent centers personal AI evolution on a correctable Personal Model rather than on transcript accumulation or skill growth alone.

understands about the person, why it believes that, what is uncertain, and what should be asked next.

We can describe the product landscape as four cumulative levels:

Level	Personal-AI capability
L1	Executes tasks: the agent follows instructions, uses tools, and delivers results.
L2	Carries context: the agent resumes a thread instead of starting from zero.
L3	Improves procedures: the agent refines skills, rules, and workflows over time.
L4	Keeps the person growing: the agent helps the user preserve judgment, evidence, questions, and growth.

Elephant Agent's product position is the fourth level. It still needs the lower levels, but it treats them as support for a more important loop: the person's understanding should deepen as the agent becomes more capable.

This distinction matters because many personal-agent systems now include memory, and many self-evolving agent systems define progress primarily as acquiring reusable skills, tools, or automations. Those capabilities are valuable, but they answer different questions. Raw recall asks what can be retrieved from the past. Skills ask what the agent can do. Personal AI also needs to answer what it currently understands about the person and how that understanding should change. Elephant Agent places this third object at the center. Memory is not discarded or narrowed to retrieval; retrieval becomes one layer in an architecture that turns episodes, relationships, risk patterns, and background reflection into correctable understanding. Capabilities can then be selected as operator-managed tools, but durable personalization remains in correctable facts, open questions, and source provenance rather than hidden skill weights or nearest-neighbor notes.

Elephant Agent implements this Personal-Model-first approach as a personal AI runtime with durable continuity. It records atomic events as Steps, binds them to execution Loops, groups Loops into Episodes, keeps Elephant State deliberately small, and evolves the Personal Model through foreground updates, proactive questions, and background reflect jobs. The clean product-facing memory architecture has five layers:

Personal Model -> Elephant State -> Step Trail -> Contextual Recall -> Background Learning

The Personal Model is the durable understanding layer. Elephant State carries the Elephant Agent identity and one natural-language current context note. An Episode is one wake or open runtime window. A Loop is one model interaction round. A Step is one atomic event inside a Loop. Together they form the Episode / Loop / Step trail. Contextual Recall retrieves support from Steps and Facts for the current turn or a reflect job. Background Learning decides when evidence should change claims or questions. Steps are the evidence; facts carry their own source Episode provenance.

This report describes Elephant Agent as a system, not as an implementation plan. The runtime provides six practical capabilities:

1. **Elephant-based continuity.** Elephant Agent resumes from a durable elephant rather than treating a session, thread, or transcript as the owner of continuity.
2. **Personal Model evolution.** Long-term user understanding is represented as active four-lens facts with confidence, source episodes, correction, and governance.
3. **Proactive curiosity.** The system can ask at quiet, balanced, or active effort levels when a gap, conflict, stale Pulse, or adaptation would materially improve future help.
4. **Claim-aware recovery.** Explicit user-directed updates change active facts in the foreground, while current-turn search can return strong, weak, or no match before Elephant Agent relies on recall.
5. **Background learning.** Feature-composable reflect agents run on Episode close, manual, diary, dream, initialization, and context-compaction triggers, then write through the same Personal Model and question tools.
6. **Capability use around understanding.** Skills, tools, models, cron jobs, messaging adapters, TUI, and Dashboard remain visible capabilities around the Personal Model.

The design goal is a personal AI that can answer four operational questions at any time: what happened, where should this line of work resume, what stable understanding exists about the user, and whether the Personal Model actually supports the claim being used now. Elephant Agent answers those questions by making persistence, provenance, and no-match recovery runtime contracts rather than prompt tricks.

The rest of the report is organized as follows. [Section 2](#) defines Personal-Model-first evolution and downstream capabilities. [Section 3](#) describes the Elephant Agent runtime architecture. [Section 4](#) defines persistence ownership. [Section 5](#) specifies facts and proactive questions. [Section 6](#) covers recall and background learning. [Section 7](#) summarizes implementation and evaluation surfaces. [Section 8](#) gives comparison axes, and [Section 9](#) outlines the next research and product directions.

2 Personal Model as the Memory Boundary

Elephant Agent treats the Personal Model as the primary object of agency-first personal AI improvement. A reusable skill can make an agent more capable, but it does not by itself say what Elephant Agent currently understands about a specific user. The Personal Model supplies that owner.

2.1 Why Skill-First Evolution Is Incomplete

Skill-first systems optimize for reusable workflows. They can discover that a tool chain often succeeds, package the path, and make it easier to invoke later. This is useful, but it leaves the core personal questions unresolved:

- What does Elephant Agent currently understand about this user?
- Which source Episodes support that understanding?
- How can the user correct, forget, or dispute it?
- Which question would materially improve future help?

These are not skill-package questions. They are Understanding System questions. Elephant Agent therefore treats skills as ordinary capabilities, while durable personalization lives in facts, questions, and provenance.

2.2 Personal Model Contents

The Personal Model is made of active facts grouped by four lenses:

Lens	Purpose
Identity	Durable attributes: name, values, decision style, boundaries, and stable self-descriptions.
World	The user's environment: people, relationships, projects, tools, domains, places, and stable context.
Pulse	Current state: active work, current pressure, recent constraints, priorities, and mood patterns.
Journey	Accumulated experience: lessons, failures, recovery patterns, past decisions, and long-running growth.

Every fact has a lens, topic, text, status, confidence, source, and `source_episode_ids`. Only active facts enter the stable prompt. Retired and disputed facts remain available for audit but must not shape answers.

2.3 Remember Less, Understand Deeper

The Personal Model is not a complete profile. It is a compact set of claims that matter for future help. The system should preserve information when it changes future collaboration, recall, risk judgment, or continuity. It should not convert every chat detail into durable truth.

This gives the product rule a technical form: remember less, understand deeper. The durable unit is not a transcript sentence. The durable unit is a correctable fact with source Episode provenance and an explicit lens/topic.

2.4 Capabilities as Downstream Tools

Skills remain useful as operator-controlled procedures. They can be installed, viewed, enabled, disabled, and invoked when relevant. They are not the owner of personal truth and are not ranked by a separate hidden affinity model.

The result is a simpler inversion: Elephant Agent does not ask only what the agent can learn to do. It asks what it currently understands about the person, why that understanding exists, and how the user can correct it.

3 Elephant Agent Runtime Architecture

The Elephant Agent runtime is organized around five cooperating memory layers: Matriarch Core / Personal Model, Elephant State, Episode / Loop / Step Trail, Contextual Recall, and Background Learning. Foreground response, explicit Personal Model updates, background reflect, and operator-managed capabilities all write through governed boundaries, and each durable update remains traceable to runtime Steps and source Episodes.

3.1 Runtime Planes

The system is easier to reason about as five planes that cooperate at runtime. The **surface plane** accepts triggers from the chat TUI, Dashboard, messaging gateways, API calls, cron jobs, and background jobs. The **response plane** owns the current foreground answer: it resolves the elephant, selects provider and model posture, calls tools, and returns a response, pause, or handoff. The **continuity plane** owns Elephant State, Episodes, Loops, and Steps, so every answer has an execution trail. The **understanding plane** owns Personal Model facts and open questions. The **learning plane** owns background LearningJobs that read Step packets and write back through Personal Model tools.

The important design decision is that these planes do not collapse into one prompt. A wake prompt is a projection of state. It can include the elephant identity, active facts, a compact Episode opening note, tool policy, and current-turn recall support. The durable owners remain the tables and tools behind those projections.

Personal-Model-First Memory Architecture

Five layers separate durable truth, state, evidence, retrieval, and learning.

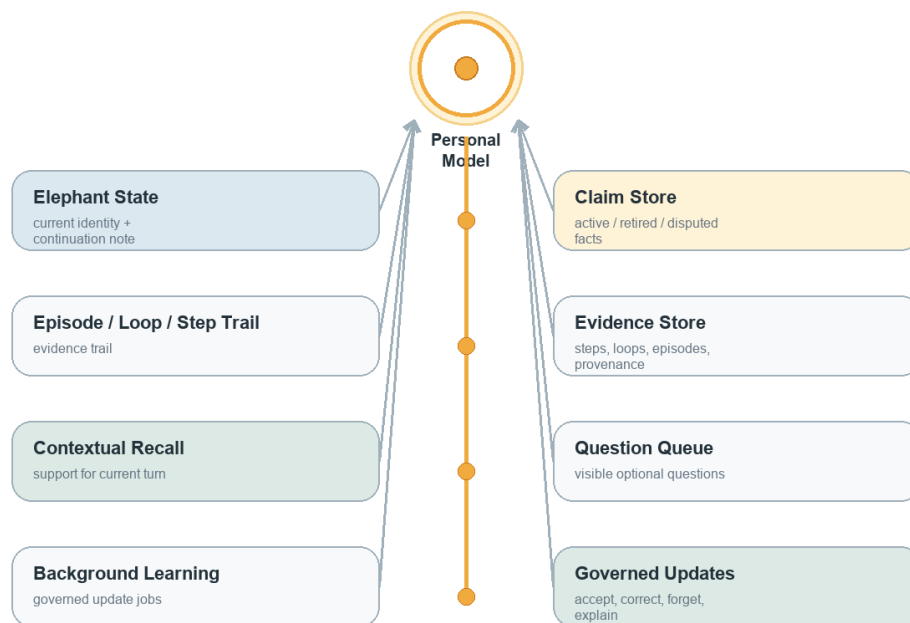


Figure 2 Elephant Agent’s Personal-Model-first memory architecture. Evidence flows upward from Step-level source material; contextual recall retrieves support for the current turn; durable claims are written only through governed Personal Model tools.

3.2 Persistence Ownership

The clean persistence model is organized around ownership boundaries, not a single memory table:

Table	Role
PersonalModel	One durable understanding container per user.
Facts	Active, retired, or disputed lens/facet claims. Facets are stable topic keys under the four lenses.
OpenQuestions	Proactive curiosity prompts and their lifecycle.
State	Elephant identity plus one current context note.
Episode / Loop / Step	Runtime windows, interaction rounds, and atomic events.
SemanticIndexEntry	Vectorized Step or Fact chunks for contextual recall.
LearningJob	Background reflect task queue and result JSON.

Steps are the canonical conversation record. Facts carry source Episode provenance.

3.3 Foreground Response

The foreground response loop is responsible for the current answer or execution. It resolves the elephant, loads active Personal Model facts and relevant State context, opens or reuses an Episode, starts a Loop, appends Steps, calls tools, and produces a response or handoff.

Prompt text is a projection of governed runtime state. It may contain active facts, an Episode opening resume snapshot, tool policy, and current-turn **Current-turn recall support**: support. It is not the owner of those facts.

3.4 Explicit Personal Model Update

When the user explicitly asks Elephant Agent to remember, correct, forget, or dispute something, the runtime treats that statement as foreground intent. The response Loop calls `tool.personal_model.update`. The update request names one lens and topic, the proposed fact text when applicable, and the reason.

Low-risk, clear, user-directed captures may commit immediately. Ambiguous, sensitive, identity-changing, cross-person, or high-impact captures require stronger policy handling. Explicit Personal Model update is allowed in the foreground because the user asked for it, but it still passes through typed policy and can retire or dispute older facts.

3.5 Claim-Aware Retrieval

Current-turn recovery searches active facts. The search surface supports strict exact lookup, conceptual semantic lookup, and verification. Exact mode disables semantic drift and accepts only strict topic, phrase, token, or high-overlap matches. Verify mode uses stricter thresholds so weak similarity is not treated as belief. Semantic mode can use translated or paraphrased query variants supplied by the model.

Every result is a claim-level answer with a status: strong match, weak match, or no match. This status is part of the runtime contract. No match is a valid answer when Elephant Agent lacks reliable Personal Model support.

3.6 Background Reflect

Inferred learning is handled outside the immediate reply. Background reflect is a feature-composable agent system. A trigger selects features, the runner builds an evidence packet from Episode Steps, the sub-agent uses foreground tools such as `personal_model.update` and `personal_model.questions`, and the runner persists its summary in `learning_jobs.result_json`.

Trigger -> features -> Step packet -> reflect agent -> tools -> result_json

Supported trigger families include `episode_close`, `manual`, `diary`, `dream`, `init_profile`, and `context_compaction`. Reflect is part of Background Learning, not a separate durable truth layer. It is a lifecycle that writes through existing tools.

3.7 Capability Path

Tools, skills, models, cron jobs, messaging adapters, the chat TUI, and the Dashboard make Elephant Agent operationally useful. They remain visible capabilities around the Personal Model. When a workflow should become reusable, it remains an explicit skill package or operator action. When the user corrects Elephant Agent, the correction updates a Personal Model fact, not a hidden skill-matching score.

4 Runtime Layers and Continuity

The Elephant Agent layer model defines what persists, for how long, and under which owner. The layers are ordered from durable understanding to atomic runtime source material.

4.1 Step

A Step is one atomic event inside a Loop. Examples include user input, model response, tool call, tool result, recall, Personal Model update, or effective user query. Steps are not durable memory owners. They are the finest-grained source material for audit, replay, recovery, debugging, and later reflection.

This distinction prevents impressionistic personalization. Elephant Agent should not learn directly from an opaque memory sentence. It should learn from structured events whose source, order, outcome, and artifacts can be inspected.

4.2 Loop

A Loop is one model interaction round. It begins with a trigger and ends when the model response, tool path, or handoff for that round is complete. A short chat reply is a Loop. A long repair path with many tools may include multiple Loops.

Runtime Layers and Continuity

A wake session carries the same thread through state, evidence, recall, and learning.

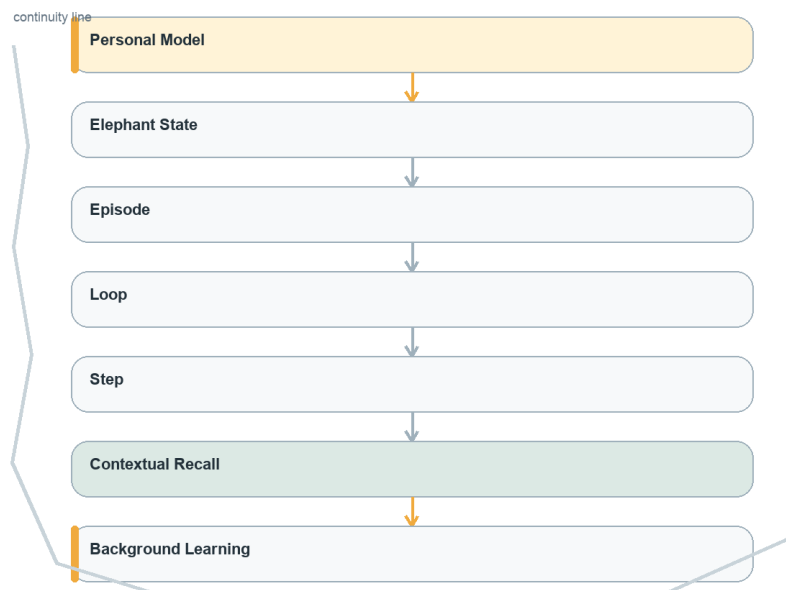


Figure 3 The Elephant Agent system layer model, shown from atomic runtime material toward durable personalized understanding.

4.3 Episode

An Episode is one open runtime window: a wake, CLI shell window, gateway worker, API handling window, or background runtime window. Ending an Episode does not end continuity. A later wake opens a new Episode and resumes from the same elephant State and Personal Model.

4.4 Elephant State

Elephant State is intentionally small:

```
elephant_id, display_name, identity_text, current_context_note, updated_at
```

`current_context_note` is the State-level continuation note produced by background Episode learning and copied into the next Episode's opening resume snapshot. It is not a per-turn working memory field and must not become a hidden task board. Live commitments belong in Episode, Step, current-turn recall, or explicit task tools.

4.5 Personal Model

The Personal Model is the durable understanding layer. It is made of active, retired, and disputed facts grouped by Identity, World, Pulse, and Journey, plus open questions that may improve future help. It is not the elephant identity itself. Deleting an elephant State does not imply deleting the Personal Model.

The simplifying rule is strict: only what should survive across multiple States, Episodes, and surfaces belongs in the Personal Model. Everything else should stay in Elephant State, Episode, Loop, Step, or current-turn recall.

4.6 Wake and Resume

Wake recovery combines Elephant State and Personal Model. Elephant Agent chooses the target elephant, opens a new Episode, loads the State resume note, projects active Personal Model facts, and may inject relevant `Current-turn recall support`: evidence for the current turn.

Proactive Curiosity Keeps the Personal Model Alive

Questions are visible, optional, user-paced, and asked only when one answer matters.

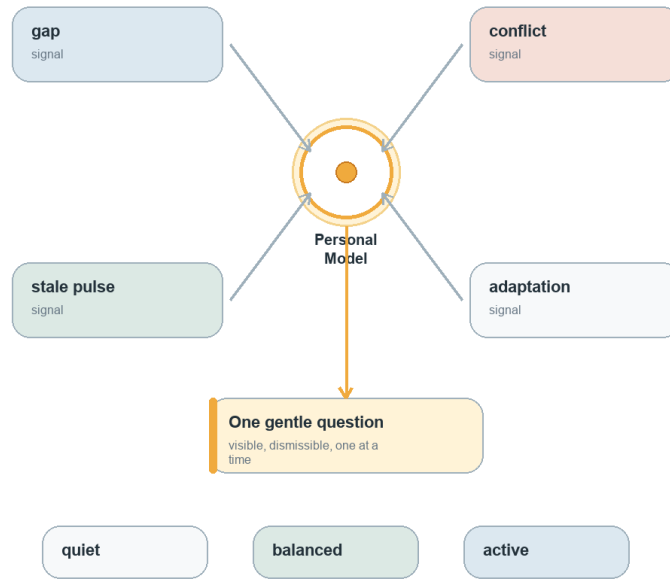


Figure 4 Proactive curiosity keeps the Personal Model alive through grounded user updates, bounded questions, background learning, and visible skill use around the model rather than hidden personality changes.

The default wake packet should answer four product questions:

1. Who is this elephant and user relationship?
2. Where did this continuity line leave off?
3. Which active facts matter now?
4. What evidence, if any, supports the current query?

This packet is not the durable owner of continuity. It is a projection. If it is wrong, the system should be able to show which facts, source Episodes, and Steps produced it.

5 Proactive Curiosity and Questions

A key claim of this paper is that a personal AI should improve by forming a small, correctable understanding of the person, not by accumulating free-form memory notes. Elephant Agent treats memory as a five-layer architecture: Personal Model, Elephant State, Episode / Loop / Step trail, Contextual Recall, and Background Learning. Within that architecture, the durable Personal Model has two objects: **Fact** and **Question**.

5.1 Facts

A **Fact** is the smallest unit of Personal Model truth. Every fact has a reference, lens, topic, text, status, confidence, source, and `source_episode_ids`. Statuses are **active**, **retired**, and **disputed**. Only active facts enter the stable prompt.

Identity Stable self-description, values, decision style, boundaries, and durable preferences.

World People, relationships, projects, tools, places, vocabulary, domains, and stable surrounding context.

Pulse Current work, current life phase, active pressure, recent constraints, priorities, and temporary mood or energy patterns.

Journey Lessons learned, past experiences, recovery patterns, repeated failures, decisions, and long-running growth trajectory.

5.2 Provenance

Provenance explains why a fact exists. It is not a separate durable storage layer. It lives in the entities that reference source material: `Fact.source_episode_ids`, Step records, and `SemanticIndexEntry` rows.

When the user asks why Elephant Agent believes something, Elephant Agent traces source Episode ids, loads relevant Steps, and presents the support material. The support can help the current turn or inspection flow, but it is not itself prompt truth.

5.3 Proactive Curiosity and Questions

A **Question** is a lens/topic-bound attempt to improve future help. Elephant Agent creates questions for four reasons only: an important gap, a conflict, a stale Pulse claim, or a high-value adaptation that would materially change future behavior. Questions are not a profile-filling checklist.

The product behavior is proactive curiosity rather than a question engine. Elephant Agent can actively maintain understanding at a user-selected effort level: *quiet* mostly waits and asks rarely, *balanced* asks at natural pauses when the answer would help, and *active* is more willing to check in while still staying optional. These levels govern frequency and willingness to communicate, not autonomous task execution.

When Elephant Agent asks a question, the lifecycle is explicit: the question moves from **open** to **asked**; the user's answer marks it **answered**; and the answer is applied through the same Personal Model update path as any other durable fact. A dismissed question stays visible as a boundary rather than being re-asked silently. Silence is honored at every effort level.

5.4 Foreground Tools

The model-facing surface is deliberately small:

- `tool.personal_model.search` reads active facts and optional provenance.
- `tool.personal_model.update` is the only foreground write path for durable understanding. It supports **remember**, **correct**, **forget**, and **dispute**.
- `tool.personal_model.questions` manages question list, ask, answer, and dismiss actions.

There is no `tool.memory.note` in the clean design. Memory is not a free-form foreground write surface. Current-turn recall is the retrieval layer; durable understanding changes only through Personal Model tools.

5.5 Prompt Projection

Stable prompt projection contains only Elephant identity, active four-lens facts, the Episode opening resume snapshot when available, and tool policy. Current-turn retrieval is injected separately as **Current-turn recall support::**; it is support for the current turn, not a source of durable truth. Retired facts, disputed facts, raw memory entries, semantic index rows, and style summaries are not projected as stable understanding.

5.6 User Visibility

The dashboard You page shows the four lenses directly. Each fact can be corrected, forgotten, or inspected through a *Why* expansion. The Questions page shows open, asked, answered, and dismissed Personal Model questions and links answered questions to the facts they produced. This makes learning inspectable: Elephant Agent does not silently model the user; it maintains a small set of facts the user can challenge.

6 Understanding Recovery and Background Learning

Continuity depends on memory, but memory is not a single retrieval primitive. Elephant Agent implements memory through five owned layers. Steps capture what happened. Elephant State keeps one continuity line coherent. The Personal Model keeps durable cross-line understanding. Contextual Recall retrieves support from Steps and Facts. Background reflect decides when Step evidence should support heavier updates.

6.1 Conversation Search

`tool.conversation.search` retrieves past conversation content for the model. It answers a different question than Personal Model search: where did this happen in the lived trail? The surface supports two modes. In `mode=discover`, it groups matching material into copyable time ranges. In `mode=recall`, it returns Step-level evidence from the selected range. It reads from:

1. **SemanticIndexEntry** for vector similarity over indexed Step or Fact chunks;
2. **Steps** for time-range fallback when semantic search is cold or empty;
3. **Episodes** for metadata aggregation in discovery mode.

Steps are the canonical conversation record. Query planning separates the topic core from recall operators such as recency, current-state, historical, recap, and verification intent. The planner handles English and CJK temporal phrases, then retrieval applies explicit windows such as `last_night`, `yesterday`, `last:3d`, date expressions, or ISO intervals. The hybrid path searches each allowed scope through vector, BM25, exact keyword, token-coverage, and n-gram signals, then reranks with a small intent-aware time score. Semantic relevance remains primary; recency only breaks near ties or helps when the user explicitly asks for recent material.

6.2 Claim-Aware Personal Model Search

When Elephant Agent needs long-term user context, it does not retrieve a generic profile blob or nearest-neighbor memory chunk. It searches active Personal Model facts. The retrieval unit is the fact, and source Steps remain support for the current turn rather than truth by themselves.

The foreground search surface combines several deterministic and semantic signals:

- topic, fact text, and source-support matches;
- Unicode lexical matching and CJK n-gram overlap;
- weak fuzzy matching for small spelling or character variation;
- semantic candidates from the durable index;
- optional translated or paraphrased query variants supplied by the model;
- lightweight polarity agreement or conflict for clear preference statements.

The result is thresholded. A search can return `strong_match`, `weak_match`, or `no_match`. This is an important safety property: if Elephant Agent cannot find a reliable active fact, it should expose the gap rather than fabricate Personal Model support.

6.3 Multilingual Hybrid Time-Aware Search

The two search surfaces share a common retrieval posture but preserve different truth semantics. Conversation search retrieves evidence from Steps and Episode summaries. Personal Model search retrieves active claims. Both rely on a multilingual hybrid search strategy:

1. **Query shaping.** The runtime normalizes Unicode text, preserves mixed-language tokens, expands CJK tokens into character n-grams, and accepts model-supplied translated or paraphrased `query_variants`.
2. **Candidate generation.** Hybrid search fuses vector similarity, BM25, token coverage, exact keyword matches, and n-gram overlap over the scoped semantic index. Personal Model search also adds fielded topic/fact matches and source-support text.

Contextual Recall Is Support, Not Truth

Retrieved evidence supports the turn; durable truth changes only through governed updates.

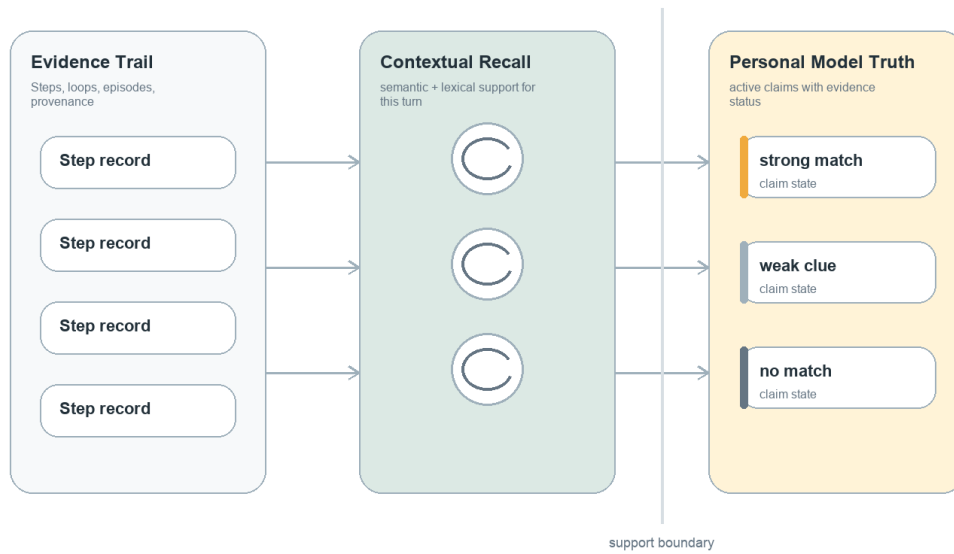


Figure 5 Contextual Recall retrieves temporary support from the evidence trail, while durable Personal Model truth changes only through governed update paths with status, confidence, and provenance.

3. **Time awareness.** Conversation search applies explicit time windows and intent-aware recency or historical reranking. Personal Model search applies volatility-based freshness so permanent claims remain stable, while situational and ephemeral claims lose rank when they have not been used recently.
4. **Judgment boundary.** Retrieval returns support, not prompt truth. Personal Model results are thresholded into `strong_match`, `weak_match`, or `no_match`; conversation results retain Episode, Loop, Step, and source-record provenance.

This lets the system recover a Chinese query over an English thread, or an English query over a Chinese claim, without relying on a global alias table or treating nearest-neighbor text as durable understanding.

6.4 Local Semantic Recall

Elephant Agent supports a local default embedding path for semantic recall. The local path lets the user run claim and conversation retrieval without requiring an external embedding provider. When desired, the embedding provider can be overridden with a configured remote endpoint and dimensions.

The default local provider is `elephant-local-embed`. It is backed by `elephant-embeddings-v1-text-small`, distributed as `llm-semantic-router/elephant-embeddings-v1-text-small` on Hugging Face and `agentic-intelligence-lab/elephant-embeddings-v1-text-small` on ModelScope. The runtime loads the model from the local Elephant Agent model root with `SentenceTransformer(..., local_files_only=True)`.

The provider exposes three online dimensions: 64, 256, and 768. A lower dimension gives faster everyday recall; a higher dimension carries more semantic detail for deeper search. The implementation normalizes vectors and uses sentence-transformers' truncation path for the selected dimension, so the same local model can support multiple latency/depth postures.

This is a support-model capability, not the durable truth layer. Embeddings help find candidate Steps or Facts. The final Personal Model truth remains the active fact with its status, lens, topic, confidence, and source Episode ids.

Multilingual Hybrid Time-Aware Search

Signals are fused into evidence-backed support, not blind transcript recall.

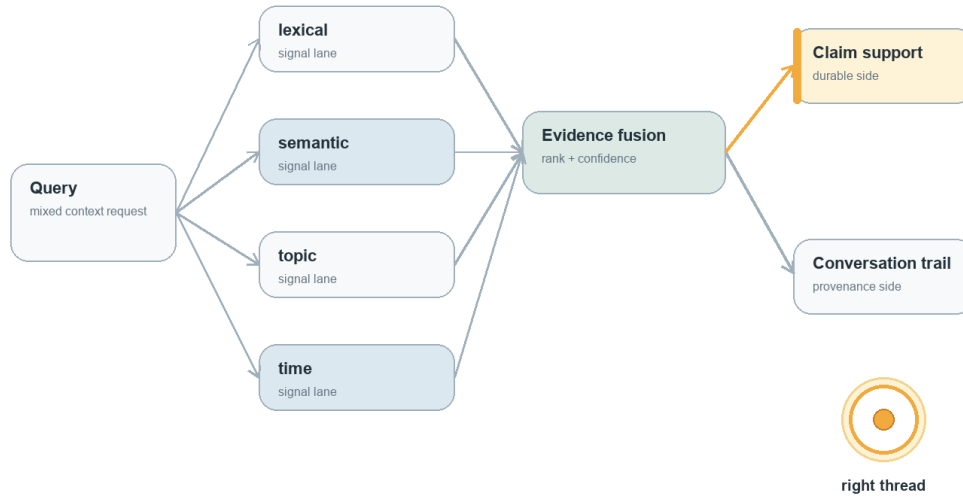


Figure 6 Multilingual hybrid time-aware search separates query shaping, evidence fusion, and temporal judgment before returning either Personal Model claim support or Conversation trail provenance.

6.5 Background Reflect

Background reflect is a feature-composable agent system. A trigger fires, a feature set is resolved, the runner composes a prompt and tools, the sub-agent receives an evidence packet built from Episode Steps, and the agent calls tools directly:

```
episode_close -> pm/questions/recall -> personal_model.update
```

Feature groups include:

- **pm**: search and write Personal Model facts;
- **questions**: create, settle, or dismiss questions;
- **recall**: search conversation history for support;
- **diary**: write reflective daily entries;
- **skills**: audit visible skill fit without creating hidden Personal Model truth;
- **compress**: check compressed content for data loss.

Reflect does not use intermediate observations, proposals, or groundings. It reads Steps directly, writes facts or questions directly through tools, and persists its run summary in `learning_jobs.result_json`.

6.6 Deletion, Correction, and Repair

Durable understanding increases the need for repair. Elephant Agent treats correction as a normal operation, not an exceptional failure. Corrections can retire active facts, dispute claims until the user clarifies them, or replace an older fact for the same lens/topic.

Persistence is valuable only when the user can inspect and correct what persists. Elephant Agent therefore treats repair paths as part of the runtime contract.

7 Implementation and Evaluation

Elephant Agent is implemented around the clean Understanding System boundary. The important implementation property is not that every old object is preserved; it is that each durable behavior has a single owner and a single correction path.

7.1 Runtime Records

The runtime separates Episodes, Loops, and Steps from durable understanding. Episodes mark open runtime windows. Loops record interaction rounds. Steps record atomic events for audit and replay. The Personal Model stores active facts and questions. Elephant State stores the elephant identity plus one natural-language current context note.

This separation lets the system compact without losing causal structure. A long tool sequence can be summarized at the Episode level while preserving critical Step links. A Personal Model fact can be corrected without mutating the transcript or treating support material as truth.

7.2 Foreground Update Path

All foreground durable understanding changes go through `tool.personal_model.update`. The tool writes one lens/topic fact at a time and supports four actions: remember, correct, forget, and dispute. Init answers, dashboard corrections, question answers, and chat-time user corrections all converge on the same path.

7.3 Claim-Aware Recall and Prompt Projection

Current-turn retrieval is contextual recall, one layer of the broader memory architecture. It is injected as **Current-turn recall support**: when useful and omitted otherwise. Stable prompt projection contains active facts only, grouped by Identity, World, Pulse, and Journey, plus the Elephant identity and Episode opening resume snapshot.

The implementation treats Personal Model search as claim-aware retrieval. The search tool supports `topic`, `query_variants`, and `mode=auto|inventory`. It merges fielded topic/fact/support matches, Unicode lexical and CJK n-gram signals, weak fuzzy matches, semantic candidates, and lightweight polarity agreement or conflict. Results are thresholded into `strong_match`, `weak_match`, or `no_match`; diagnostics can expose per-fact signals for evaluation but are not prompt truth.

Topics are represented as facets: stable keys under the four lenses that attach each claim to one slice of understanding without adding a second profile schema.

`tool.conversation.search` provides the complementary evidence path. A small deterministic query planner strips recall operators from the topic core and detects recency, current-state, historical, recap, and verification intent across English and CJK phrasing. `mode=discover` requires a time range and returns candidate windows; `mode=recall` retrieves Step or Episode evidence from a selected window. The searcher first attempts the durable semantic index with vector, BM25, exact keyword, token-coverage, and n-gram signals, then falls back to Step scanning when the index is cold. Final ranking keeps semantic relevance primary and uses temporal intent only as a small recency or historical tie-breaker.

7.4 Operational Evaluation

Elephant Agent is evaluated through longitudinal scenarios rather than single-turn prompt quality alone:

Scenario	Expected behavior
Wake recovery	Resume an episode using the Elephant context note, active facts, and relevant recall.
Claim correction	Replace an active fact through <code>correct</code> , retire the old fact, and project only the new fact next turn.
Forget	Retire a fact through <code>forget</code> so it no longer shapes replies.
Question answer	Mark the question answered and create the resulting fact with source Episode provenance.
Why inspection	Show source Steps behind a fact without promoting those Steps to prompt truth.

Scenario	Expected behavior
Search no-match	Return no active fact, with a reason, when query quality or scores are too weak to support an answer.
Verify mode	Require stronger support before treating a retrieved fact as support for a proposed statement.
Local recall	Retrieve relevant Step or Fact chunks through the local embedding path without requiring external embedding services.
Multilingual search	Recover relevant claims or Step evidence when the query and stored material use different languages or mixed-language phrasing.
Time-aware search	Respect explicit time windows and temporal intent without letting recency override stronger semantic or lexical evidence.
Local inference posture	Verify that <code>elephant-local-embed</code> can steady, index, and retrieve at 64, 256, and 768 dimensions with the same Step/Fact corpus.
Background reflect	Run a feature-scoped learning job and persist its summary in <code>learning-jobs.result_json</code> .

The evaluation question is whether Elephant Agent can continue the right work with the right current understanding, and whether the user can correct that understanding when it is wrong.

7.5 Case Study Shapes

The most informative case studies are multi-session and multi-surface:

- a repository refactor where Elephant Agent resumes with the correct Elephant context and recalls only relevant support;
- a design discussion where a correction to collaboration style changes the next reply;
- a dashboard fact correction where the old fact is retired and the new fact appears in the prompt;
- an answered Personal Model question that becomes an Identity, World, Pulse, or Journey fact with inspectable provenance;
- a recall query that correctly returns `no_match` instead of inventing support;
- a background diary or Episode-close job that updates the Personal Model through tools and stores a learning result.

The reporting unit is the complete lifecycle: trigger, recalled support, runtime action, fact update, provenance path, dashboard visibility, and later wake behavior.

8 Related Systems and Comparison Axes

This report does not add unverified citations. A citation-verified related systems pass should be added later through programmatic reference checks rather than memory-based BibTeX entries. The comparison axes are still clear. They also map to the four-level product view used in the public README and website: execution, context, procedure improvement, and human growth.

- **Stateless chat systems.** These optimize the current prompt but do not preserve durable collaboration state. They are closest to L1 task execution.
- **Long-context systems.** These extend the input window but do not by themselves define what should become State or Personal Model truth. They help with L2 context but do not settle ownership.
- **Memory-augmented assistants.** These retrieve prior snippets, but often leave explicit capture, inferred personalization, maturity, correction, and scope boundaries underspecified.
- **Workflow agents.** These execute tool-heavy tasks, but often treat persistence as logs and checkpoints rather than as a layered personal model.
- **Companion systems.** These move closer to Elephant Agent when they model the user, relationship, and elephant identity boundaries, but they often blur Episode, State, Personal Model, and procedural learning.

Memory Evaluation Lifecycle

A memory is useful when it improves future judgment and can be corrected.

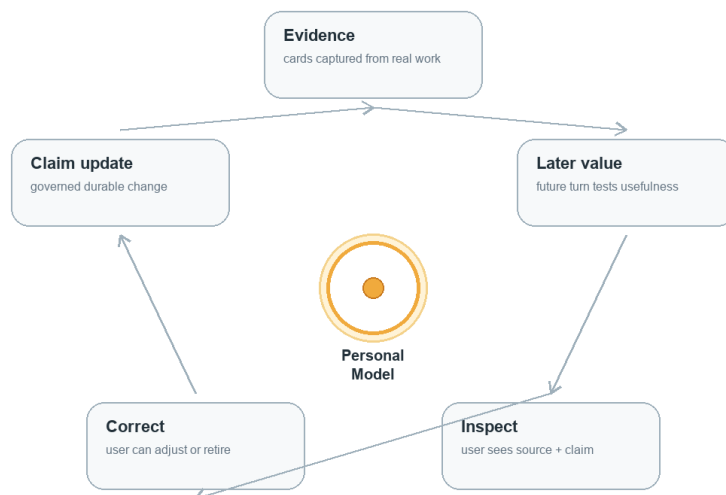


Figure 7 Memory evaluation should measure the full lifecycle: evidence, retrieval, claim update, user correction, dashboard inspection, and later wake behavior, with future work focused on local inference quality, independent memory evaluation, and co-design with routing and inference systems.

- **Skill and automation systems.** These make repeated workflows reusable, but they rarely define the boundary between durable personal understanding and executable skill artifacts. They are closest to L3 procedure improvement.
- **Skill-first self-evolving agents.** These optimize for acquiring reusable skills, tools, or workflows. Elephant Agent treats that as useful capability work, but not as the center of personal AI. The center is the Personal Model: active facts, source provenance, and useful questions.
- **Agency-first personal AI.** This is the L4 axis. The question is whether the product keeps judgment, evidence, questions, and growth with the person while lower-level agent capabilities become cheaper and more powerful.

Elephant Agent is therefore best compared on ownership boundaries. Atomic events belong in Steps. Execution belongs in Loops. Wake windows belong in Episodes. Workline continuity belongs in State. Long-term personalization belongs in the Personal Model. Background reflect turns Step source material into fact changes and question candidates without collapsing learned truth into a skill package.

9 Future Work

Elephant Agent opens five concrete research and product directions.

First, local inference should become faster and more semantically reliable. The current local path uses a compact sentence-transformers model with 64/256/768 dimensional retrieval modes. Future work should improve latency, multilingual hybrid recall quality, time-aware ranking calibration, stance-aware matching, claim-local query expansion, and robustness under long-running local indexes while keeping active prompt truth in Personal Model facts.

Second, memory evaluation should become independent of generic chat quality. Useful benchmarks should measure wake recovery, correction latency, Personal Model maturity, no-match calibration, social

and risk recall, background learning precision, and whether later answers improve without overriding user corrections. The evaluation unit should be a lifecycle: evidence, retrieval, claim update, user correction, dashboard visibility, and later wake behavior.

Third, background skill learning should grow beyond matching estimation. Skill matching is only the first signal: it tells the system which existing skills fit the user’s work and style. A mature background learner should also notice repeated workflows in the Step trail, propose new user-owned skills when history reveals a stable pattern, and suggest refinements to existing skills when evidence shows that a procedure is brittle, stale, or missing local context. This should remain governed by provenance and operator visibility: the system may propose or improve skill material, but durable skills should be inspectable and correctable rather than hidden adaptations.

Fourth, the security boundary should become a first-class research surface. Personal-model-first agents combine long-lived understanding with tools, code execution, messaging bridges, and background jobs. Future work should harden coding sandbox isolation, tool access control, permission scoping, secret handling, audit trails, and policy checks that decide which tools a foreground or background agent may call. The important unit is not only a single tool invocation, but the whole path from recalled support to model decision to tool execution to durable write.

Fifth, memory should be co-designed with routing, inference, and reinforcement learning rather than optimized as an isolated retrieval feature. The end-to-end path is:

agent -> router -> inference -> train

The routing layer should learn when a turn needs Personal Model search, conversation search, local embedding, a stronger model, a safer model, or no recall at all. This naturally connects to vLLM Semantic Router: route by memory need, language, latency, privacy posture, jailbreak risk, tool-access risk, confidence, and model-collaboration shape rather than by model cost alone. It also connects to the vLLM inference engine: local and served inference should expose latency, batching, cache, and KV-cache optimization signals so Elephant Agent can choose the right recall and generation posture per turn.

The training side of this co-design is RL post-training over the full agent lifecycle. The target is not only better tool use or higher task completion. Training signals should reward asking at the right time, refusing weak memory support, preserving user corrections, choosing when to write or not write a claim, selecting the right search surface, using skills only when they fit the current Personal Model and consent context, and routing through the right model and inference posture for the user’s privacy and safety constraints.

10 Conclusion

Elephant Agent implements agency-first personal AI through Personal-Model-first memory. The system does not treat reusable skill acquisition or raw recall as the primary improvement target. It treats the Personal Model as the center: the durable, grounded model of the user across Identity, World, Pulse, and Journey.

The clean Understanding System gives that center a concrete path:

Personal Model -> Elephant State -> Step Trail -> Contextual Recall -> Background Learning

The Personal Model stores active facts and open questions. Elephant State keeps identity and one current context note. The Episode / Loop / Step trail records atomic source material. Contextual Recall retrieves support without turning it into prompt truth. Background Learning reads Steps directly, writes through the same foreground tools, and persists learning results for inspection.

This structure changes how personal AI improves. Resuming is not transcript replay, and learning is not only adding reusable skills. Instead, explicit Personal Model updates are controlled, proactive curiosity operates at the user’s chosen effort level, inferred personalization matures through reflection, and current-turn recall searches active claims with no-match as a safe outcome. The result is a fourth-level product posture: lower-level execution, context, and procedure improvement remain useful, but they serve the user’s judgment, evidence, questions, and growth.

The practical outcome is a personal AI that can explain what happened, continue the right line of work, update long-term understanding through governed provenance, repair itself after correction, and use capabilities while keeping durable understanding inspectable. Skills still matter. In Elephant Agent, they remain downstream operator-managed tools around an evolving Personal Model.